



DuoMix Fader

Play two apps at once, mix them with a crossfader — and see, while the music plays, which scale it is in and which chords it loops through.

User guide · version 0.9 · September 2026

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- | | |
|---|--|
| <ul style="list-style-type: none">1 What it does2 Getting started3 The mixer4 The fader in your notifications5 Harmony: scale, tonic, chords6 What Harmony listens to: app sound or microphone | <ul style="list-style-type: none">7 History, one section per piece8 The Scales of Harmonies sheet9 How the analysis works, and its limits10 Phones and Android versions11 Privacy12 Credits |
|---|--|

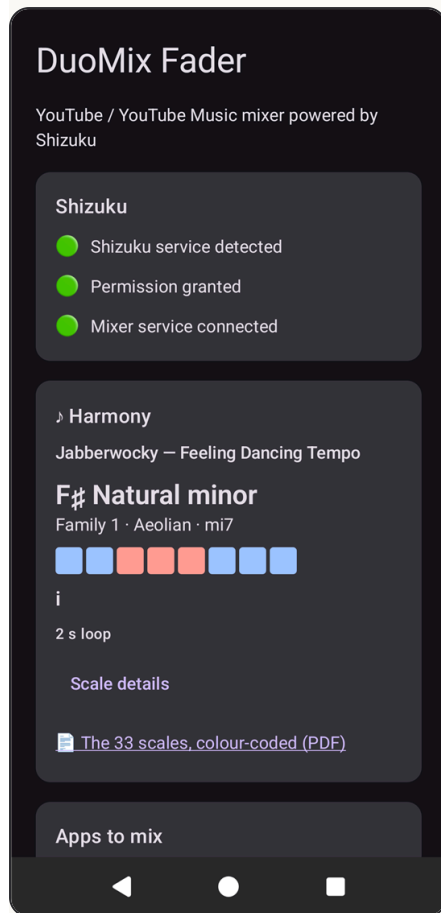
1 • What it does

Android normally lets one app play at a time: start a video and your music pauses. DuoMix Fader removes that limit for two apps you choose — a **music** app and a **video or voice** app — and gives you a two-channel mixer with a DJ-style, constant-power crossfader. It needs no root.

On top of the mixer, the **Harmony** feature listens to the music channel and estimates the scale of the piece among the 7 families and 33 *Scales of Harmonies* described by Oliver Prehn (NewJazz), its tonic, and its chord progression in Roman numerals. It is an accompaniment aid: a steady, averaged reading of the piece, not a note-by-note transcription.

DuoMix Fader is an independent project. It is not affiliated with Google, YouTube, the Shizuku project or Oliver Prehn.

2 • Getting started



Three green lights: Shizuku is running, permission is granted, the mixer service is connected.

DuoMix Fader relies on **Shizuku**, a separate free app that lets it perform a few system-level audio operations without root.

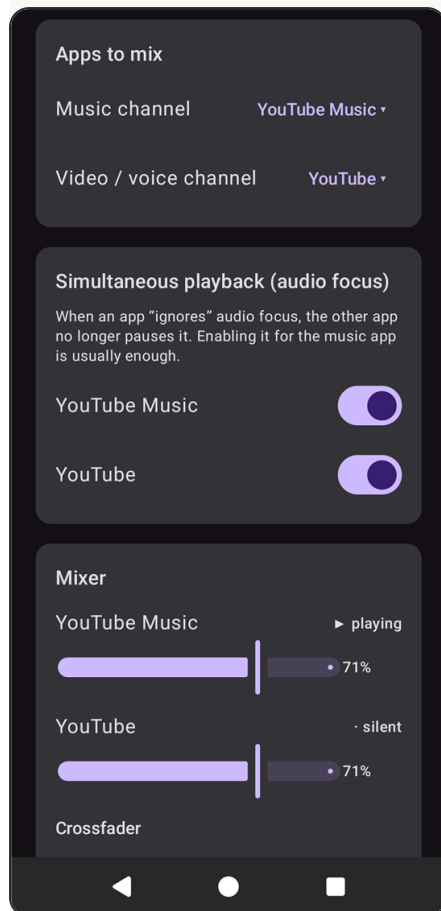
1. Install **Shizuku** and start it through *Developer options* → *Wireless debugging*, following Shizuku's own instructions. After a reboot it has to be started again, unless you enabled its start-on-boot option.
2. Open **DuoMix Fader** and tap **Request Shizuku permission**, then accept.
3. When the three lights are green, you are ready.

If the first light is red, Shizuku is not running: open Shizuku and start it. Nothing in DuoMix Fader works without it. On a **OnePlus, OPPO or realme** phone, or on **Android 12**, read section 10 first.

Languages

The app follows your phone's language — English, French or Spanish. You can also choose its language alone in *Settings* → *System* → *Languages* → *App languages*. Scale names are international musical terms and are never translated.

3 • The mixer



Apps to mix, the audio-focus switches, the two channel sliders and the crossfader.

Apps to mix

Pick one app per channel. Only supported apps that are installed are listed: for music, YouTube Music, Spotify, Deezer, Apple Music, Amazon Music, SoundCloud, Tidal, Qobuz — and YouTube, X and Telegram, since music is also played from videos and feeds (Harmony listens to the music channel); for video and voice, YouTube, X, Telegram, Twitch, VLC, Netflix, Pocket Casts, Chrome, Firefox. An app can hold only one channel: pick it on a channel while the other one has it, and the two channels swap. The list is closed on purpose: the privileged service refuses any other app.

Simultaneous playback

Android pauses an app when another one takes the *audio focus*. Turn the switch on for an app and it stops being paused by the other. Enabling it for the **music app** is usually enough.

Some apps pause *themselves* when they are refused audio focus (X does). Their switch is disabled and says so: enable the option on the other channel instead.

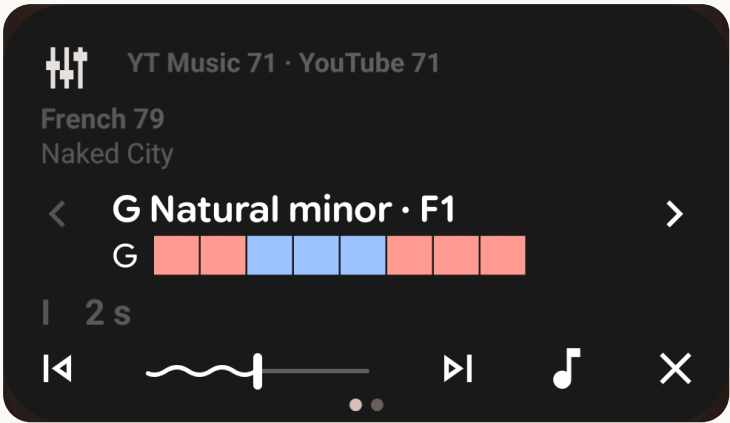
Volumes and crossfader

Each slider sets the volume of one app only. The crossfader moves from the music app (left) to the video app (right) following an equal-power curve: at the centre both sit at 71 %, so the overall loudness stays constant.

When you close both the screen and the notification, both apps get their full volume back, and switching a channel to another app restores the app you leave.

4 • The fader in your notifications

You do not have to keep the app open. While DuoMix Fader runs, a media-player card called **DuoMix fader** appears next to your other players — in the notification shade and on the lock screen.



The card, cropped from the notification shade. Here: G natural minor, family 1 — G A in red, B $\flat$ C D in blue, E $\flat$ F G in red — a few seconds into a piece, when the progression is still a single chord. The name of the audio output is masked.

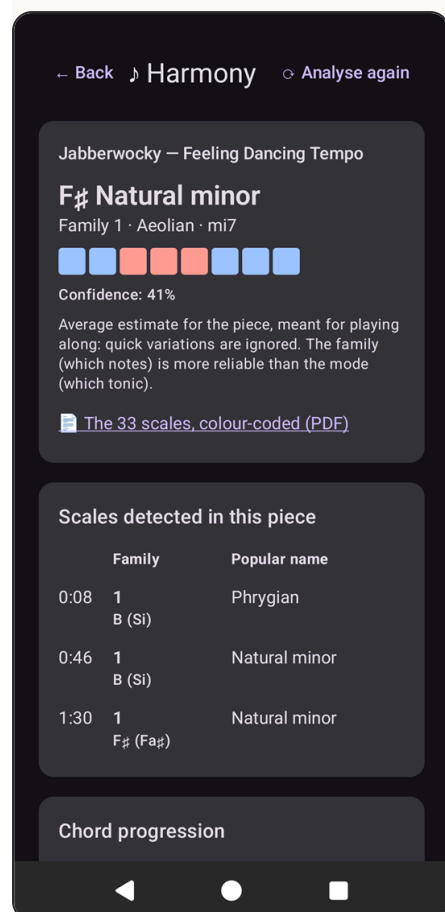
ON THE CARD	WHAT IT SHOWS OR DOES
Title	The essential: the mode. Tonic, popular name and family — “E Mixolydian · F1”.
Colour tiles	The key, in letter notation — “A”, “D $\flat$ ” — then one tile per note of the scale, from the tonic to its octave . Blue and red are the two whole-tone sets of the keyboard, so a change of colour means a half step or a step and a half. A minor reads R R B B B R R R. This sequence is the fingerprint of the mode. A tile  split in two is reached by a step and a half (families 3, 4 and 7): the colour you come from on top, the one you arrive at below.
Chord progression	Under the tiles, in Roman numerals relative to the tonic: “i – bVI – iv – bVI 8 s” when a loop was found, with its length; “... i – iv – bVII” — the last chords heard — when none was. It only changes once a new progression has held for four seconds.
Small text, top	To the right of the app icon, the two apps and their volumes: “YT Music 71 · YouTube 71”. Beneath, on two lines, the artist and the title of the piece; a long title is cut short. This text is drawn in the card's background artwork, which Android darkens.
Progress bar	It is the crossfader: drag it. Left is the music app, right is the video app. To re-centre, drag it to the middle.
⏮ / ⏭	Move the fader 10 % towards music / towards video.
🎵	Close the shade and open the Harmony panel.
✕	Hide the card. It comes back the next time you open the app.

DuoMix Fader plays no sound itself: this player card is only a control surface, which is why it has no play button. Its layout, its two text lines and their size belong to Android. Nothing scrolls or blinks on the card: it only changes when the scale, the piece or the balance does.

The history widget

Long-press an empty spot of your home screen, choose *Widgets*, then **DuoMix • Detection history**. It shows the pieces you listened to — the one playing first — each with its key, its scale and colour tiles, the scales held in turn with their degree, and the chord progression. Scroll the list; tap it to open the history in the app. It is updated only when its content changes, never on a timer, and it is not offered on the lock screen: the player card is there for that.

5 • Harmony: scale, tonic, chords



The piece as announced by the music app, the scale found, the scales held during the piece, and the chord progression.

Open it from the **Harmony** card of the main screen (*Scale details*) or from the **Harmony** button of the notification. Analysis starts by itself as soon as the music app plays.

What you read

- **The scale** — tonic and popular name, then its family, systematic name and chords, as in Oliver Prehn's table. The tonic is written in your language's system, with the other one next to it: "B $\flat$ (Si $\flat$)".
- **Confidence** — how clearly this set of notes beats the next candidate. Below about 15 %, treat the family as a hint.
- **Scales detected in this piece** — one row per scale held, with the time it appeared and the five columns of the table. Scroll the table sideways.
- **Chord progression** — Roman numerals relative to the tonic, the chord names beneath, and the length of the loop.

Reading the numerals

Degrees are read on the **major scale of the tonic**, with $\flat$ or $\sharp$ for other pitches: in A minor, F major is written $\flat$ VI. Upper case is major (+ for augmented), lower case is minor ($^\circ$ for diminished).

Analyse again

The **Analyse again** button forgets everything and analyses the current piece afresh — useful if you started listening in the middle of an unusual passage.

The reading is deliberately **slow and steady**. The first estimate needs about 8 seconds of music; a change of scale is only accepted if it lasts 12 seconds, a change of mode 30 seconds. Expect the first minute of a piece to be the least reliable.

6 • What Harmony listens to: app sound or microphone

Under the scale, the line **Source** offers two buttons. You can switch at any time; the analysis, the history, the widget and the notification card follow whichever source is active.

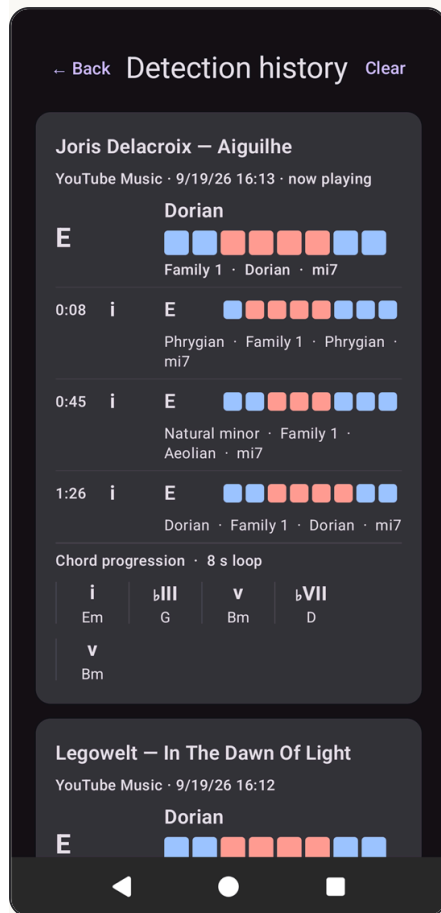
SOURCE	WHAT IT HEARS
App sound	The sound of the music app only , taken inside the phone: it works with headphones, and the other channel never enters the analysis. On recent versions of Android the privileged service captures it. On Android 12 it cannot, so DuoMix Fader uses Android's own <i>playback capture</i> instead: the first time, Android asks for the recording permission — it is used only to read the music app's sound, the screen is never captured — and may show a cast icon in the status bar while it runs.
🔊 Microphone	What the phone hears : its own speaker, a stereo in the room, an instrument you play. It hears nothing of a headphone session. Tap it once to start (Android asks for the microphone permission the first time), tap it again to stop. It is never switched on by itself.

Microphone settings

- **Gain**, 0 to +40 dB. A phone microphone is quiet when it is read without processing, and music below a certain level is treated as silence. Raise the gain until the scope comes alive.
- **Bass** and **Treble**, –12 to +12 dB. Lifting the bass helps the bass line and the tonic; lowering the treble sets room noise aside. At 0 dB the sound is left untouched.
- **The scope** shows three lanes — treble, mids, bass — taken *after* these settings, at a fixed scale. Flat traces: raise the gain. Traces shaved off at the top and bottom: it clips, lower it. Waves filling about half of each lane on loud passages are about right.

When the microphone is off **and** the music app is switched off under *Simultaneous playback*, nothing is listened to: the block says **Not listening**. Turn one of the two back on. An app whose switch cannot be turned on (X) is still analysed.

7 • History, one section per piece



One section per piece: main scale, scales held in turn with their degree, chord progression.

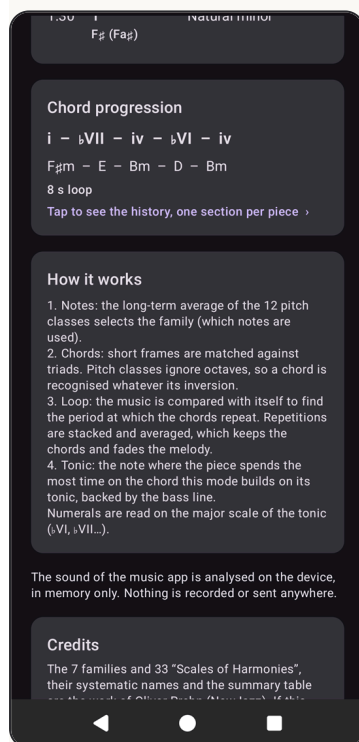
Tap the **Chord progression** card of the Harmony panel to open the history. The piece being played comes first, marked *now playing*; earlier pieces follow, most recent first, up to fifty.

Each section reads in three levels, separated by thin lines:

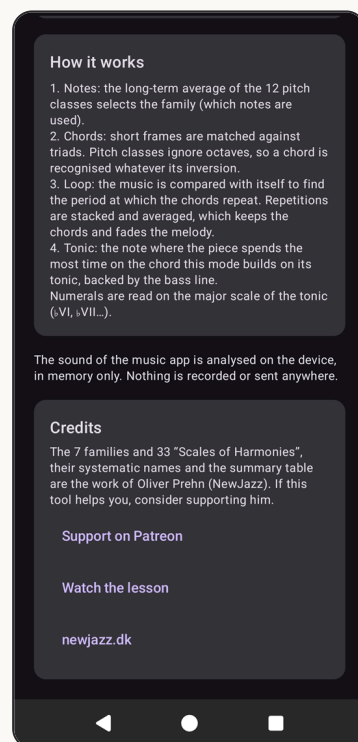
- **The main scale** of the piece, under its name: the key in large letters, the scale and its **colour tiles**.
- **The scales held in turn**, one row each, in fixed columns: the time, the **degree in Roman numerals** relative to the main key, the tonic, the tiles. Read down a column to follow the move from one degree to the next.
- **The chord progression**, at the foot: one cell per chord, the degree above its name, and the length of the loop.

Pieces are told apart by the title and artist that the music app announces itself — the same information your lock screen shows. Nothing is recognised by listening and nothing is looked up online. When an app announces no title, a silence of a few seconds marks the boundary and the section is called *Unknown piece*.

Clear empties the list. The history is a private file of the app; it contains names of pieces, scales and chords, never any sound.



The panel also explains the method in a few lines, and states what happens to the sound.



Credits, with links to Oliver Prehn's lesson, website and Patreon page.

8 • The Scales of Harmonies sheet

The link  **The 33 scales, colour-coded (PDF)**, in the Harmony card and in the panel, opens a one-page study sheet inside the app. Pinch to zoom, drag to move around. It exists in English, French and Spanish and follows the app's language.

The Scales of Harmonies

An overview of 33 scales in 7 families, by **Oliver Prehn** (NewJazz) — colour-coded study sheet

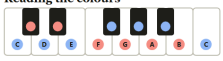
The 33 scales presented in the NewJazz lesson are organised in 7 families. Each family has one interval pattern; a scale is that pattern read from a given degree — which is why the rows of a family line up as a staircase. The 33 scales follow from a few simple rules:

- Only half, whole and whole-and-a-half
- Half steps may not be neighbours
- Whole and whole-and-a-half steps may not be neighbours
- Whole-and-a-half steps may not be neighbours

Following them limits the total to 33 scales, all well suited to building clear, evident harmonies.

Lesson: youtu.be/Vq2xtD3e3E • www.newjazz.dk • Support Oliver Prehn: patreon.com/newjazz • paypal.me/newjazz

Reading the colours



Blue and red are the two whole-tone sets of the keyboard. A whole step (1) stays in the same colour; a half step (1/2) or a step and a half (1 1/2) crosses to the other. Each scale is a row of tiles, one per note, from the tonic to its octave: that two-colour sequence is the fingerprint of the mode. Colours are shown with the first scale of each family starting on C; down a family, each column is one and the same note. From a red tonic, swap the two colours. A tile split in two is reached by a step and a half: the colour you come from on top, the one you arrive at below — not to be mistaken for a half step (families 3, 4 and 7).

FAMILY	POPULAR NAME	INTERVAL STEPS	*SYSTEMATIC NAME	CHORDS
1	Major	1-1-1/2-1-1-1-1-1/2	Ionian	Ma7
	Dorian	1-1/2-1-1-1-1/2-1	Dorian	m17
	Phrygian	1/2-1-1-1-1/2-1-1	Phrygian	m17
	Lydian	1-1-1-1/2-1-1-1/2	Lydian	Ma7
	Mixolydian	1-1-1/2-1-1-1/2-1	Mixolydian	7
	Natural minor	1-1/2-1-1-1/2-1-1	Aeolian	m17
	Locrian	1/2-1-1-1/2-1-1-1	Locrian	m17b5
2	Altered / Super Locrian	1/2-1-1/2-1-1-1-1	Ionian #1	7alt / m17b5
	Ascending mel. minor	1-1/2-1-1-1-1-1/2	Dorian #7	m1Ma7
	Dorian b2	1/2-1-1-1-1-1/2-1	Phrygian #6	m17
	Lydian Augmented	1-1-1-1-1/2-1-1/2	Lydian #5	Ma7#5
	Lydian dominant	1-1-1-1/2-1-1/2-1	Mixolydian #4	7
	Aeolian dominant	1-1-1/2-1-1/2-1-1	Aeolian #3	7
	Half diminished	1-1/2-1-1/2-1-1-1	Locrian #2	m17b5
3	Major #5 / Major Aug.	1-1-1/2-1/2-1/2-1-1/2	Ionian #5	Ma7#5
	Dorian #4	1-1/2-1/2-1/2-1-1/2-1	Dorian #4	m17
	Phrygian dominant	1/2-1/2-1/2-1-1/2-1-1	Phrygian #3	7
	Lydian #2	1/2-1/2-1-1/2-1-1-1/2	Lydian #2	Ma7
	Altered dominant bb7	1-1/2-1/2-1-1-1/2-1/2	Mixolydian #1	d1m7
	Harmonic minor	1-1/2-1-1-1/2-1/2-1/2	Aeolian #7	m1Ma7
	Locrian #6	1/2-1-1-1/2-1/2-1/2-1	Locrian #6	m17b5
4	Harmonic Major	1-1-1/2-1-1/2-1/2-1/2	Ionian b6	Ma7
	Dorian b5	1-1/2-1-1/2-1/2-1/2-1	Dorian b5	m17b5
	Phrygian b4	1/2-1-1/2-1/2-1/2-1-1	Phrygian b4	m17 / 7
	Lydian b3	1-1/2-1/2-1/2-1-1-1/2	Lydian b3	m1Ma7
	Mixolydian b2	1/2-1/2-1/2-1-1-1/2-1	Mixolydian b2	7
	Lydian augmented #2	1/2-1/2-1-1-1/2-1-1/2	Aeolian b1	Ma7#5 / d1m7
	Locrian bb7	1/2-1-1/2-1-1/2-1/2-1/2	Locrian b7	d1m7
5	Diminished	1-1/2-1-1/2-1-1/2-1-1/2	Diminished	d1m7
6	Dominant diminished	1/2-1-1/2-1-1/2-1-1/2-1	Inverted diminished	7
7	Whole tone	1-1-1-1-1-1-1	Whole tone	7 #5/b5
	Augmented	1/2-1/2-1/2-1/2-1/2-1/2	Augmented	Ma7
	Inverted Augmented	1/2-1/2-1/2-1/2-1/2-1/2	Inverted Augmented	6#5

*The systematic names do not follow the jazz-school habit of naming every scale after the Major scale. *Aeolian #7 is an Aeolian scale with a raised 7th step: Aeolian is the premise, #7 the alteration. The first 4 families thus run from Ionian to Locrian with a simple sequence of sharps or flats.

Musical content © Oliver Prehn, reproduced for study with credit; all rights remain with its author. Layout and colour code: Steve Nodcock, DuoMia Fader, 2026. Handwritten notes on the source sheet are not included.

The sheet reproduces the data of Oliver Prehn's overview — 33 scales in 7 families, with popular name, interval steps, systematic name and chords — and adds a colour code:

-  **Blue** and  **red** are the two whole-tone sets of the keyboard: C D E F# G# A# and C# D# F G A B.
- Each scale is a **row of square tiles**, one per note, from the tonic to its octave — the same notation as the notification card and the history. A **whole step** stays in the same colour; a **half step** or a **step and a half** crosses to the other.
-  A tile **split in two** is reached by a **step and a half**: the colour you come from on top, the one you arrive at below. A plain change of colour is a half step. Only families 3, 4 and 7 contain steps and a half. The same mark is used in the app, the widget and the notification.
- Colours are given with the first scale of each family starting on C. The scales of a family line up as a **staircase** — each one is the family's pattern read from the next degree — so every column is one and the same note. From a red tonic (C#, F, G, A...), swap the two colours: the app always shows the real ones.
- Each family has its own colour; the scales printed in bold in the original are bold here too.

The musical content of this sheet is Oliver Prehn's work and remains his. Only the layout and the colour code were added. See *Credits*.

9 • How the analysis works, and its limits

1. **Notes** → **family**. The energy of the 12 pitch classes is averaged over about thirty seconds. The set of notes that explains it best gives the family. When two candidates are close, the more common family wins.
2. **Chords**. Short frames are matched against the 48 triads. Pitch classes ignore octaves, so a chord is recognised whatever its inversion or voicing.
3. **Loop**. The music is compared with itself to find the period at which the chords come back. All repetitions are stacked and averaged: what returns every time — the chords — is reinforced; melody, voice and drums fade. If no loop is found, the last 20 seconds are read as they are, and the panel says so.
4. **Tonic** → **mode**. The modes of a family share the same notes, so notes alone cannot tell D Dorian from C major. The tonic is the note where the piece spends the most time on the chord that this mode builds on its first degree — the “Chords” column of the table — backed by the bass line.

Limits you should know

- **The mode is less certain than the family**, and sometimes the ambiguity is musical: a loop such as I – V – vi – IV spends as much time on four chords, each of which is the tonic chord of a mode of the same family.
- Rootless jazz voicings can hide a chord (a rootless Cmaj9 is an Em7); the bass line is then the only clue.
- Chords shorter than about half a second, music tuned far from A = 440 Hz, and very percussive mixes are read poorly. Four-note chords are reduced to their triad.
- If the music channel is faded to zero, there is nothing left to analyse.
- The method has been tested on synthetic signals and on a handful of real pieces, not on an annotated corpus: its success rate on real music is unknown.

The full method, its constants and what did not work are described in the research note shipped with the source code, `docs/research/harmony-analysis.md`.

10 • Phones and Android versions

DuoMix Fader acts through the rights that Android gives to its *shell* identity, and those rights are not the same on every version or brand. The app checks what your phone allows when it starts, switches off what it cannot do, and says so on the screen. It installs from Android 12.

	RECENT ANDROID TESTED: PIXEL 11 PRO XL, ANDROID 17	ANDROID 12 TESTED: ONEPLUS 7 PRO, EMULATOR
Two apps playing at once	Yes	Yes
Per-app volume, smooth crossfade	Yes	No — switch mode instead
Harmony on the app's sound	Yes, by the privileged service	Yes, by Android's playback capture
Harmony by the microphone	Yes	Yes
Notification	A media-player card: fader, buttons, scale, tiles, progression	An ordinary notification drawn by the app (see below)

Switch mode

Android 12 does not let the shell reach the players of other apps, so their volume cannot be set. The shell can still switch an app's sound off and on. The fader then becomes a three-way switch — **music alone** on the left, **both** in the middle, **video alone** on the right — and each channel slider mutes its app below 40 %. The mixer card says *Switch mode* and shows each channel as *on* or *muted*. Whatever was muted gets its sound back when DuoMix Fader stops, and again the next time it starts, should it ever be killed in between.

The notification on Android 12

The media-player card of Android 12 cannot show a scale: its text line lets only two tiles through, and it shrinks the artwork to a small square. There, DuoMix Fader posts an ordinary notification instead, outside the media area. **Collapsed**, it shows the key and the eight tiles in a row. **Expanded**, it adds the name of the mode, the chord progression, the piece, and the fader as three buttons — music app, *Both*, video app, the current one outlined — next to ♪ and ✕.

OnePlus, OPPO and realme

These brands remove some rights from adb — and therefore from Shizuku — until you turn on **Disable permission monitoring** in *Developer options*, then restart Shizuku. Without it Shizuku warns that “the permission of adb is limited” and DuoMix Fader cannot work; the app reminds you of the option. It lifts a protection that these brands add on top of Android for a computer connected in debugging mode, and gives adb back the behaviour it has on other phones: decide for yourself whether that suits the phone you use.

Versions between Android 13 and 16, and other brands, have not been tested. The app does not guess from the version number: it reads what the shell is allowed to do on the phone in hand.

11 • Privacy

- The app has **no network permission**. Nothing it hears, reads or stores can leave the phone.
- To analyse a piece it listens to the **music app only** — or, if *you* choose it, to the phone's **microphone**. Either way the sound is reduced on the spot to twelve numbers per instant (and, for the microphone, to the picture of the scope) and discarded; it is never recorded, and the microphone is never switched on by itself.
- The **recording permission** is asked only when you choose the microphone, or *App sound* on Android 12, where Android's playback capture requires it. There, the capture is limited to the music app's sound; the screen is never captured.
- It reads the title and artist that the music app publishes for the lock screen, to name the sections of the history.
- The history is a private file; **Clear** deletes it. Uninstalling the app removes everything.
- The privileged service it starts through Shizuku can only do a short, fixed list of things — change the audio-focus setting and the volume of the listed apps (or, on Android 12, switch their sound off and on), listen to the music app, read its title, open DuoMix Fader's own screen, and allow DuoMix Fader itself to use Android's playback capture. It runs no arbitrary command and stops by itself when the app stops.

12 • Credits

The 7 families and 33 *Scales of Harmonies*, their systematic names and the overview table are the work of **Oliver Prehn** (NewJazz). If this tool helps you, please support him.

- Lesson: [ALL ABOUT MUSICAL SCALES – A COMPLETE GUIDE!!!](#) — youtu.be/Vq2xt2D3e3E
- Website: www.newjazz.dk · Support: patreon.com/newjazz · paypal.me/newjazz

His material is not covered by the app's open-source licence and all rights remain with him. The detection method, the app, the colour code and this guide are by Steve Nodock, with Claude (Anthropic) for algorithms and implementation; they are released under the Apache License 2.0. Shizuku is by RikkaApps.